

# **BUSINESS PROBLEMS**

## **The Master Bank of 25 Cybersecurity Business Real Life Problems**

### **Phase 1: Macro Baselines & Historical Trends**

- **Problem 1 (Baseline Frequency):**  
Which industry had the most successful cyber-attacks, what was the average SOC response time for those attacks, and what was the total volume of data compromised (GB)?
  - **Problem 2 (The Macro Trend):**  
Which historical year in our dataset experienced the absolute highest volume of attacks, and which year experienced the absolute least?
  - **Problem 3 (The 2020s Outlook):**  
Looking across the multi-year epoch of the 2020s so far, which specific global country has recorded the least number of total attacks?
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### **Phase 2: Geographic & Chronological Anomalies**

- **Problem 4 (The Runner-Up Exception):**  
Isolating our data strictly to the United Kingdom, which specific security incident ranks as the *second* most severe attack on record?
  - **Problem 5 (The 2023 Serious Impact):**  
Which specific cyber-attack had the most serious, multi-variable impact (evaluating severity scale, duration, and data loss) during the calendar year 2023?
  - **Problem 6 (The 2020 Perimeter Trend):**  
Which exact calendar month in the year 2020 recorded the absolute lowest frequency of malicious activity originating from blocked IP addresses?
  - **Problem 7 (The US Same-Day Anomaly):**  
In the United States, within which specific month and hour did a high-severity, high-data-loss ("brutal") attack strike, yet our defences managed to completely resolve it on the exact same calendar day?
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### Phase 3: SLA Breach Audits & Operational Friction

- **Problem 8 (The Tool Efficiency & Latency Gap):**

For all successful breaches where the 60-minute SLA was violated, what is the matrix breakdown of Security Tool Used paired with the Mitigation Method deployed? Which combination represents our highest financial exposure?

- **Problem 9 (The Extended Blast Radius):**

What is the total count of catastrophic incidents where *both* the Attack Duration exceeded 120 minutes *and* the SOC Response Time breached the 60-minute SLA? Which Attack Type caused the most data loss under these conditions?

- **Problem 10 (The Data Exfiltration Velocity Rate):**

When an attack is marked as successful, what is the exact difference in the average volume of Data Compromised (GB) for incidents resolved *within* the 60-minute SLA versus those that breached it?

- **Problem 11 (The Low-Volume SLA Trap):**

For the global country identified as having the least attacks in the 2020s, what is our actual SLA compliance rate? What percentage of those few attacks still breached our 60-minute response protocol?

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### Phase 4: Human Risk & Privilege Escalation

- **Problem 12 (The Inside Threat Temporal Cluster):**

When high-privilege accounts (Admin or Executive) are compromised, what is the exact correlation between the Incident Hour of the attack and our SOC Response Time? Are SLA breaches clustered around specific shifts?

- **Problem 13 (The 2022 Operational Friction):**

In the year 2022, when we look at attacks that fell into the critical Need Action duration tier ( <250 minutes), how were they distributed across different User Roles? Were high-privilege accounts disproportionately stuck in this slow-response bucket?

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## Phase 5: Custom Feature Engineering & Categorization

- **Problem 14 (Duration Tier Bucketing):**

For the year 2022, write the conditional logic to categorize and count all attacks into four operational speed zones based on their duration:

- *Quick*: < 50 minutes
- *Normal*: 50 - 150 minutes
- *Slow*: 150 - 250 minutes
- *Need Action*: <250 minutes

- **Problem 15 (The High-Damage Swift Resolution):**

Which specific industry suffered a maximum-severity, high-data-loss ("brutal") attack, yet the incident was successfully routed and resolved in the *Quick* duration tier (< 50minutes)? What month did this occur?

- **Problem 16 (The Ghost Hunt / False Alarms):**

Which Attack Type accounts for the highest volume of incidents where the severity was negligible ( $\leq 3$ ) and data loss was 0 GB, but our SOC team still took over 60 minutes to resolve it, needlessly triggering an SLA penalty?

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## Phase 6: Operational Anomalies & Efficiency Benchmarks

- **Problem 17 (The OS Perimeter Breakdown):**

Within the United States perimeter for 2023, which specific Security Tool Used had the highest failure rate (Outcome = 'Success')? When these tools failed, did a specific operating system environment suffer the brunt of the damage?

- **Problem 18 (The Persistent Threat Vector):**

For all high-severity attacks striking US infrastructure, what is the frequency distribution of the Mitigation Method used? How often did teams rely on manual IP Blocking versus automated Patch Management, and what was the response time gap between them?

- **Problem 19 (The Asset Vulnerability Matrix):**

When a cyber-attack successfully breaches our 60-minute SLA threshold in the United States, which specific Target System (e.g., Database, Cloud Storage, Infrastructure) is compromised most frequently, and what is the primary threat vector used against it?

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## Phase 7: Final Data Validation & Audit Triggers

- **Problem 20 (The Elite Regional Benchmark):**

Which global country represents the absolute gold standard of defensive operations by maintaining both the *least* total number of attacks *and* resolving them in the *Quick* duration tier?

- **Problem 21 (The Elite Blueprint Blueprint):**

For the champion country identified in Problem 20, what is their exact underlying infrastructure profile? What specific distribution of security tools and mitigation methods do they utilize to achieve such high-efficiency containment?

- **Problem 22 (The Data Integrity & Cross-Validation Audit):**

Before publishing our final executive dashboard, identify any data anomalies where an attack log claims the Outcome = 'Failed' (the attack was blocked) but the row records a value greater than 0 for Data Compromised (GB) or lists an active Attack Duration.

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## Phase 8: Future Prediction

- **Problem 23 (The (APT) Low-and-Slow Attack Chain):**

Identify any stealth threat actors by finding instances where the same Attacker IP targeted the exact same Target System across multiple separate incidents, where each individual attack had a low severity score ( $\leq 3$ ) and lasted less than 30 minutes, but the aggregate time gap between the first and last incident spanned more than 14 days.

- **Problem 24 (The Cascading Infrastructure Failure & Blast Radius Propagation):**

Find catastrophic "burst windows" where a successful breach on a critical system triggered a chain reaction, defined as an initial high-severity attack ( $=10$ ) on a Target System followed by at least three subsequent successful attacks on *different* systems within the exact same country inside a tight 6-hour rolling time window.

- **Problem 25 (The SOC Operational Fatigue & Burnout Efficiency Decay Trend):**

Calculate the running 7-day moving average of our SOC Response Time grouped by engineer shift patterns to determine if a massive spike in total incident volume over a consecutive 48-hour period directly causes a statistically significant decay (slowdown) in operational response efficiency over the subsequent 5 days.